

Explore the Benefits of Data Storage Systems



Data storage systems are used to store, access and safeguard data and enable its efficient management. They facilitate quick and efficient data retrieval. Learn more about them in this article.

As I write this, I am fondly reminded of my first computer, and how I loved the fact that I had one! I could program. I could watch movies and burn CDs to store GBs of data like software trials, Linux and the e-books I gathered from friends. That was the time I spent almost entire Sundays on maintenance. One of the things that enthralled me was that I could store almost 20 movies on my computer, apart from all the software installations. There was a HDD that could store 40 GB of data, and all I could say about those 40 GB back then was, “Wow!”

Today, I have over 3 TB of storage space. I have a number of multimedia files and I don't really care where they are, and there is hardly a day I spend on maintenance. In the last eight years, we have seen a lot of changes in the field of computing. One of the biggest has been in the amount of data we can store and process every day. The likes of Facebook, Google and Dropbox have changed the way we treat data. Gone are the days when you wondered about how to rack up the 1000 pictures that Orkut said it would let you store. Facebook says it has hundreds of billions of pictures, and those numbers are constantly increasing.

Storing, managing, searching and processing enormous amounts of data are some of the prime challenges that computing science has been trying to solve. The good thing is that we seem to be succeeding at it.

What are data storage systems?

The question, in its simplest form, answers itself—a system that allows you to store data can be called a data storage system. So if you have a computer at home on which you have kept all your data, the computer is acting as your data storage system. A network share that stores files is one of the very basic data storage systems.

Where are data storage systems used?

As I mentioned earlier, a network share is good enough to be called a data storage system. But let's look further. High-end data storage systems of today are designed to store petabytes of data and are mostly distributed. How else would you store the index of the entire Web (think Google) or store more than half the pictures taken by all mankind, through the history of this planet (think Facebook). So as far as usage goes, I guess it's clear by now that data storage systems are used to store almost all kinds of data - from simple network shares to data stores that allow one to save and retrieve data ranging from a collection of pictures to the complex index of the entire Web.

Types and use-cases

Before you go ahead, let me mention that details on how to set up any of the software discussed below is beyond the scope of this article.